Logistics regression – Mathematics

Linear combination:

$$z = m_1x_1 + m_2x_2 \ + \ \ldots \ + \ m_nx_n + c$$

The sigmoid function:

$$g(z) = \frac{1}{1 + e^{-z}}$$

The loss function:

The requirement is to have a higher loss the further away the value is from the true label and a lower loss the closer the value is from the true label

$$L(g(z_i), y_i) = \begin{cases} -\log(g(z_i)) & \text{if } y_i = 1 \\ -\log(1 - g(z_i)) & \text{if } y_i = 0 \end{cases}$$

Combining this into one expression:

$$L(g(z_i), y_i) = -y_i\log(g(z_i)) - (1 - y_i)\log(1 - g(z_i))$$

The cost function:

$$\mathbf{C} = \frac{1}{n}\sum_{i=1}^{n} L(g(z_i), y_i)$$

$$\mathbf{C} = \frac{1}{n}\sum_{i=1}^{n}(-y_i\log(g(z_i)) - (1 - y_i)\log(1 - g(z_i)))$$

Finding the partial derivatives of the cost function:

$$\frac{\partial L}{\partial m_j} = \frac{\partial L}{\partial g_i} \times \frac{\partial g_i}{\partial m_j} = \frac{\partial L}{\partial g_i} \times \frac{\partial g_i}{\partial z_i} \times \frac{\partial z_i}{\partial m_j}$$

$$= \left(\frac{g_i - y_i}{g_i(1 - g_i)} \right) \times g_i(1 - g_i) \times x_{i,j}$$

$$= (g_i - y_i)(x_{i,j})$$

$$\Rightarrow \frac{\partial C}{\partial m_j} = \frac{1}{n} \sum_{i=1}^{n} (g_i - y_i)(x_{i,j})$$

$$\frac{\partial L}{\partial c} = \frac{\partial L}{\partial g_i} \times \frac{\partial g_i}{\partial c} = \frac{\partial L}{\partial g_i} \times \frac{\partial g_i}{\partial z_i} \times \frac{\partial z_i}{\partial c}$$

$$= \left(\frac{g_i - y_i}{g_i(1 - g_i)} \right) \times g_i(1 - g_i) \times 1$$

$$= g_i - y_i$$

$$\Rightarrow \frac{\partial C}{\partial c} = \frac{1}{n} \sum_{i=1}^{n} (g_i - y_i)$$

Formula used in the gradient descent algorithm:

$$m = m - \alpha \left(\frac{\partial C}{\partial m} \right)$$

$$c = c - \alpha \left(\frac{\partial C}{\partial c} \right)$$

The confusion matrix:

$$\begin{bmatrix} \textbf{True zeros} & \textbf{False ones} \\ \textbf{False zeros} & \textbf{True ones} \end{bmatrix}$$

This is a table that shows how well a classification model's predictions match the actual answers. I will use it to find the precision and recall.

Precision = Of the wines I predicted as good, how many were actually good?

Recall = Of all the actually good wines, how many did I correctly find?

$$\text{Precision} = \frac{\textbf{True ones}}{\textbf{True ones} + \textbf{False ones}} \times 100$$

$$\text{Recall} = \frac{\textbf{True ones}}{\textbf{True ones} + \textbf{False zeros}} \times 100$$

Standardization:

$$z = \frac{X - \mu}{\sigma}$$